A. Feder Cooper, Ph.D.

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I research a variety of topics in reliable, scalable machine learning. My contributions span privacy, security, evaluation of generative-AI systems, MLSys, and measurement validity. I also do work in tech policy and law, and spend a lot of time finding ways to effectively communicate the capabilities and limits of AI/ML to interdisciplinary audiences and the public. My research has received spotlights, orals, and best-paper accolades at top AI/ML and computing venues, including *NeurIPS*, *ICML*, *AAAI*, and *AIES*. Collaborations on copyright and generative AI have been lauded as "landmark" work among scholars and the international press.

Current Position

Assistant Professor, Department of Computer Science, Yale University
Co-lead, The GenLaw Center
Founding Research Scientist, AI Verification and Evaluation Research Institute (AVERI)
Affiliate Researcher, Stanford HAI, RegLab, and CRFM
Faculty Associate at the Berkman Klein Center for Internet & Society at Harvard University

Education

Cornell University
Ph.D., Computer Science 2018 – 2024
M.S., Computer Science 2021

Columbia University
B.A., Computer Science and Archaeology, Phi Beta Kappa, *summa cum laude* 2014

Selected Papers

*Denotes co-first author or equal contribution; †lead author (not specified if same as first author); **Google Scholar**

A. Feder Cooper, Marika Swanberg, Jamie Hayes, Lea Duesterwald, Christopher De Sa, Daniel E. Ho, Mark A. Lemley, and Percy Liang. "Extractable Memorization From First Principles." 2026. [**link**] **arXiv**

Mark A. Lemley* and **A. Feder Cooper***. "Probabilistic 'Copies' in Generative AI Models." Forthcoming, *Berkeley Technology Law Journal*. 2026. [**link**] **Journal**

A. Feder Cooper, Mark A. Lemley, Christopher De Sa, Daniel E. Ho, Percy Liang, and others. "Extracting memorized pieces of (copyrighted) books from open-weight language models." *COLM '26*. [**link**] **Proceedings** (29% acceptance; written 2025; *ICML* Workshop **Oral**)

A. Feder Cooper, Mark A. Lemley, Christopher De Sa, Lea Duesterwald, Allison Casasola, Jamie Hayes, Katherine Lee, Daniel E. Ho, and Percy Liang. "Estimating near-verbatim extraction risk in language models with decoding-constrained beam search." *COLM '26*. [**link**] **Proceedings** (29% acceptance; *ICML* Workshop **Oral**)

Miles Brundage, Noemi Dreksler, Aidan Homewood, Sean McGregor, Patricia Paskov, Conrad Stosz, Girish Sastry, **A. Feder Cooper**, and others. "Frontier AI Auditing: Toward Rigorous Third-Party Assessment of Safety and Security Practices at Leading AI Companies." 2026. [**link**] **arXiv**

Ahmed Ahmed*, **A. Feder Cooper***, Sanmi Koyejo, and Percy Liang. "Extracting books from production language models." 2026. [**link**] **arXiv**

Jamie Hayes*, **A. Feder Cooper***, Ilia Shumailov, Christopher A. Choquette-Choo, Matthew Jagielski, Milad Nasr, Katherine Lee, and others. "Exploring the limits of strong membership inference attacks on large language models." *NeurIPS '25*. [**link**] **Proceedings** (24.52% acceptance)

A. Feder Cooper*†, Katherine Lee*†, Miranda Bogen*, James Grimmelmann, danah boyd, Percy Liang, Daniel E. Ho, Pamela Samuelson, Amy B. Cyphert, Mark A. Lemley, Nicolas Papernot, and others. "Machine Unlearning Doesn't Do What You Think: Lessons for Generative AI Policy and Research." *NeurIPS '25, Position Track*. [link] **Proceedings** **Oral** (≈0.4% of submitted papers).

Jamie Hayes*, Marika Swanberg, Harsh Chaudhari, Itay Yona, Ilia Shumailov, Milad Nasr, Christopher A. Choquette-Choo, Katherine Lee, and **A. Feder Cooper***. "Measuring memorization in language models via probabilistic extraction." *NAACL '25*. [link] **Proceedings** (∼22% acceptance)

A. Feder Cooper* and James Grimmelmann*. "The Files are in the Computer: On Copyright, Memorization, and Generative AI." *Chicago-Kent Law Review*, Vol. 100, 2025. [link] **Journal** (written 2024)

Katherine Lee*, **A. Feder Cooper***, and James Grimmelmann*. "Talkin' 'Bout AI Generation: Copyright and the Generative-AI Supply Chain." *Journal of the Copyright Society*, Volume 72, 2025. [link] **Journal** (written 2023)

Nicholas Carlini, Daniel Paleka, Krishnamurthy Dj Dvijotham, Thomas Steinke, Jonathan Hayase, **A. Feder Cooper**, Katherine Lee, Matthew Jagielski, Milad Nasr, Arthur Conmy, Eric Wallace, David Rolnick, and Florian Tramèr. "Stealing Part of a Production Language Model." *ICML '24*. [link] **Proceedings** **Best Paper** (≈0.1% of submitted papers)

A. Feder Cooper, Solon Barocas, Christopher De Sa, Jon Kleinberg, and others. "Arbitrariness and Social Prediction: The Confounding Role of Variance in Fair Classification." *AAAI '24*. [link] **Proceedings** **Best Student Paper Honorable Mention** (<1% of submitted papers; written 2023)

A. Feder Cooper*†, Wentao Guo*, Khiem Pham*, and others. "Coordinating Distributed Example Orders for Provably Accelerated Training." *NeurIPS '23*. [link] **Proceedings** (26% acceptance)

Milad Nasr*, Nicholas Carlini*, Jonathan Hayase, Matthew Jagielski, **A. Feder Cooper**, Daphne Ippolito, Christopher A. Choquette-Choo, Eric Wallace, Florian Tramèr, and Katherine Lee†. "Scalable Extraction of Training Data from (Production) Language Models." 2023. [link] **arXiv**

A. Feder Cooper, Jonathan Frankle, and Christopher De Sa. "Non-Determinism and the Lawlessness of Machine Learning Code." *CSLAW '22*. [link] **Proceedings**

A. Feder Cooper*†, Emanuel Moss*, Benjamin Laufer, and Helen Nissenbaum†. "Accountability in an Algorithmic Society: Relationality, Responsibility, and Robustness in Machine Learning." *FAccT '22*. [link] **Proceedings** (25% acceptance)

A. Feder Cooper, Yucheng Lu, Jessica Zosa Forde, and Christopher De Sa. "Hyperparameter Optimization Is Deceiving Us, and How to Stop It." *NeurIPS '21*. [link] **Proceedings** (<26% acceptance)

A. Feder Cooper and Ellen Abrams. "Emergent Unfairness in Algorithmic Fairness-Accuracy Trade-Off Research." *AIES '21*. [link] **Proceedings** **Oral** (< 10% of submitted papers)

Ruqi Zhang*, **A. Feder Cooper***, and Christopher De Sa†. "Asymptotically Optimal Exact Minibatch Metropolis-Hastings." *NeurIPS '20*. [link] **Proceedings** **Spotlight** (<3% of submitted papers)

Selected Honors, Grants, and Awards

Stanford HAI Seed Grant ($75,000), "Assessing Copyright Risks of Training Data 'Memorization' and 'Extraction' for Open-Weight Language Models" (2026 – 2027)

Meta Ph.D. Fellowship Finalist (2023) (short-listed out of 3200 candidates)

Rising Star in EECS, MIT (2021)

Fellowship Finalist, OpenPhil AI Ph.D. Fellowship (2021)

Selected Conference, Workshop, and Tutorial Organization

Data Privacy, Memorization, & Legal Implications in Generative AI: A Practical Guide. *NeurIPS '25* Tutorial, December 2, 2025 (San Diego, CA).

2nd Workshop on Generative AI and Law (GenLaw '24). Held at *ICML 2024* in Vienna, Austria. Lead co-organizer with Katherine Lee. [**link**]

Evaluating Generative AI Systems: the Good, the Bad, and the Hype (GenLaw DC '24). Lead co-organizer. Sponsored by the K&L Gates initiative in Ethics and Computational Technologies at Carnegie Mellon University (CMU), and organized by the GenLaw Center (A. Feder Cooper, Katherine Lee, James Grimmelmann), Carnegie Mellon's K&L Gates Initiative (Hoda Heidari), the Georgetown Law Center (Paul Ohm), and the Center for Democracy and Technology (Alexandra Reeve Givens, Miranda Bogen). [**link**]

1st Workshop on Generative AI and Law (GenLaw '23). Held at *ICML 2023* in Honolulu, Hawai'i, 26% acceptance rate. Lead co-organizer with Katherine Lee; co-organized with Dr. Niloofar Mireshghallah, Madiha Zahrah Choksi, Prof. James Grimmelmann, Prof. David Mimno, and Dr. Deep Ganguli [**link**]

Selected Recent Talks

Invited participant, Symposium on AI and Law, Harvard Law School, October 24, 2025 (Cambridge, MA).

"What Copyright Can Learn from Memorization Measurements of Language Models."

- Invited talk, American National Standards Institute (ANSI), May 11, 2026 (Virtual).
- Invited speaker, Workshop & Lecture Series on the Law & Economics of Innovation, ETH Zürich, October 28–29, 2025 (Zürich, Switzerland).
- Invited speaker, Information Technology and Innovation Foundation, August 2025 (Virtual).
- Invited speaker, **Workshop on the Impact of Memorization on Trustworthy Foundation Models**, *International Conference on Machine Learning*, July 2025 (Vancouver, Canada).

Invited panelist, **Workshop on Machine Unlearning for Generative AI**, *International Conference on Machine Learning*, July 2025 (Vancouver, Canada).

AI Governance Tutorial (with Katherine Lee, 28th Annual BTLJ-BCLT Spring Symposium: AI Governance at the Crossroads, February 27, 2025 (Berkeley, CA).

"Machine Unlearning Doesn't Do What You Think: Lessons for Generative AI Policy, Research, and Practice."

- Future of Privacy Forum, Technologist Roundtable for Policymakers, July 2025 (Virtual).
- Invited talk, NYU Law Innovation Policy Colloquium, April 3, 2025 (New York, NY).